

Restaurant Management System Database Project

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1. Introduction

The Restaurant Management System (RMS) database is developed to manage all restaurant operations in an organized and structured way. This database integrates customers, reservations, orders, employees, chefs, inventory, suppliers, payments, billing, and managerial supervision into one centralized system.

The main purpose of this system is to maintain data consistency, eliminate redundancy, ensure referential integrity, and improve operational efficiency. The ER diagram represents the logical structure of the database and shows how entities interact with each other to maintain smooth workflow inside the restaurant.

This system supports daily operations such as order processing, table reservations, billing generation, and stock monitoring. It also improves customer satisfaction by maintaining accurate records and fast service management.

2. System Description

The system consists of multiple interconnected entities. Each entity represents a specific component of restaurant operations. Customers can make reservations and place orders. Orders are handled by employees and include multiple order items.

Each order item refers to a menu item, and each menu item is prepared by a chef based on specialization. Payments are recorded separately and generate bills including tax calculations.

Inventory is supplied by suppliers and managed under the supervision of the manager.

This ensures accountability and stock control. The system is designed to be scalable and can be implemented using relational database systems such as MySQL or PostgreSQL.

3. Uniqueness of the Database Design

This database design is unique because it connects the entire restaurant workflow from customer entry to final billing. The reservation system is directly linked with table management, preventing double booking.

Orders are connected with employees, enabling performance tracking. Menu items are linked to chefs, ensuring specialization accountability. Payment and billing are separated to allow proper financial reporting and tax calculation.

Future enhancements can include loyalty programs, online ordering systems, real-time inventory alerts, feedback systems, and AI-based demand prediction. These features make the system modern and adaptable.

4. Entities and Attributes

CUSTOMER: customer_id (PK), name, email, phone_number, address.

RESERVATION: reservation_id (PK), date, time, number_of_people, customer_id (FK), table_id (FK).

TABLES: table_id (PK), table_no, capacity, status.

ORDER: order_id (PK), date, time, type, status, customer_id (FK), employee_id (FK), table_id (FK).

ORDER_ITEM: order_item_id (PK), order_id (FK), menu_item_id (FK), quantity, subtotal.

MENU_ITEM: menu_item_id (PK), item_name, category, price, availability, chef_id (FK).

CHEF: chef_id (PK), name, phone, speciality, salary.

EMPLOYEE: employee_id (PK), name, position, phone, salary.

PAYMENT: payment_id (PK), order_id (FK), method, amount, date.

BILL: bill_id (PK), payment_id (FK), total_amount, tax, date.

SUPPLIER: supplier_id (PK), name, contact, address.

INVENTORY: inventory_id (PK), item_name, unit, quantity,
expiry_date, supplier_id (FK), manager_id (FK).

MANAGER: manager_id (PK), name, phone, salary.

5. Relationships in the Database

Customer – Reservation: One-to-Many

Customer – Order: One-to-Many

Reservation – Tables: Many-to-One

Order – Order_Item: One-to-Many

Order_Item – Menu_Item: Many-to-One

Menu_Item – Chef: Many-to-One

Order – Employee: Many-to-One

Order – Payment: One-to-One

Payment – Bill: One-to-Many

Supplier – Inventory: One-to-Many

Inventory – Manager: Many-to-One

These relationships ensure referential integrity and structured data connectivity.

RELATIONS IN RESTAURANT MANAGEMENT SYSTEM

No.	Relationship	From Entity	To Entity	Cardinality
1.	MAKES	CUSTOMER	RESERVATION	1 : M (One to Many)
2.	PLACE	CUSTOMER	ORDER	1 : M (One to Many)
3.	ASSIGNED	RESERVATION	TABLES	M : 1 (Many to One)
4.	HAS	ORDER	PAYMENT	1 : M (One to Many)
5.	GENERATE	PAYMENT	BILL	M : 1 (Many to One)
6.	GIVEN TO	BILL	SUPPLIER	1 : 1 (One to One)
7.	SUPPLIES	SUPPLIER	INVENTORY	1 : M (One to Many)
8.	MANAGED BY	INVENTORY	MANAGER	M : 1 (Many to One)
9.	CONTAIN	ORDER	ORDER ITEM	1 : M (One to Many)
10.	REFER TO	ORDER ITEM	MENU ITEM	M : 1 (Many to One)
11.	PREPARED BY	MENU ITEM	CHEF	M : 1 (Many to One)

6. Data Normalization

The database follows normalization principles up to Third Normal Form (3NF). Each table has a primary key to uniquely identify records. Foreign keys are used to maintain relationships between tables.

There are no partial dependencies or transitive dependencies. This reduces data redundancy and improves data integrity. Normalization ensures that updates and deletions do not create anomalies within the system.

7. Implementation Strategy

The system can be implemented using SQL-based relational database systems. Tables are created using CREATE TABLE statements with constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, and UNIQUE.

Indexes can be added on frequently searched fields such as order_id and customer_id

to improve performance. Role-based access control can be applied to ensure security and data protection.

8. Advantages of the System

The system improves operational efficiency, enhances customer experience, reduces manual errors, ensures financial accuracy, and provides structured inventory management. It supports business growth and scalability.

9. Outcome of the Project

The final outcome of this project is a fully structured Restaurant Management System database capable of handling all essential operations of a restaurant.

It ensures accurate order tracking, proper billing, efficient reservation management, and effective inventory supervision. The database enhances decision-making and improves overall management performance.

This system provides a strong foundation for implementing advanced features in the future and supports long-term operational success.

10. Conclusion

In conclusion, the Restaurant Management System database is a comprehensive and scalable solution designed to manage all aspects of restaurant operations. It integrates customers, staff, inventory, suppliers, and financial transactions into a unified structure.

The database ensures data accuracy, integrity, and efficiency. With further enhancements, it can evolve into a smart and intelligent restaurant system.

- **Visualization/ER**

Restaurant Management System

